Ejafa Bassam

M.Sc. in Data Science, Institute of Data Science & Engineering
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PROFILE

M.Sc. graduate in Data Science from Peking University, with research experience in learning objectives, knowledge distillation, and data augmentation. First author of papers at **NeurIPS 2025** and **NeurIPS 2026**, leading projects from mathematical formulation and implementation through large-scale empirical evaluation. My current interests center on how better supervision and training objectives can help models learn more efficiently within a given parameter and compute budget. Seeking doctoral research in deep learning or deep reinforcement learning.

EDUCATION

M.Sc. in Data Science Jul 2023 - Jul 2026
Peking University, Beijing, China
Institute of Data Science & Engineering
GPA: 3.70/4.00

B.Sc. in Computer Science Sep 2019 - Jul 2023
Jiangnan University, Wuxi, China
School of Artificial Intelligence
GPA: 3.84/4.00

PUBLICATIONS

2026 **Ejafa Bassam**, Konstantin Garbers, Yingsheng Geng, Dalton Jens, Jiaxu Qian, Guyue Liu.
TreemapMix: Dirichlet-Controlled Multi-Image Augmentation for Probability and Ordinal Supervision
Advances in Neural Information Processing Systems (NeurIPS 2026). [Link]

2025 **Ejafa Bassam**, Dawei Zhu, Kaigui Bian.
PLD: A Choice-Theoretic List-Wise Knowledge Distillation
Advances in Neural Information Processing Systems (NeurIPS 2025). [Link]

2026 Khushal Haider Syed, Ruben Poghosyan, Hongjian Shi, Ruhui Ma, Yuan Luo, **Ejafa Bassam**, Ziang Zhou, Zhongle Qu, Haibing Guan.
DecentFL: A Decentralized Federated Learning System with Multiple Aggregators and Evaluators
IEEE Internet of Things Journal, published July 6, 2026. [Link]

Academic Service
Reviewer, **NeurIPS 2026**

RESEARCH EXPERIENCE

Research Assistant Jul 2023 - Jul 2026
Institute of Data Science & Engineering
School of Computer Science, Peking University
Advisor: Dr. Kaigui Bian

TECHNICAL SKILLS

Programming: Python, C/C++, Go, Rust, Bash
ML: PyTorch, JAX, Hugging Face
Tools: Git, Linux, Docker, LaTeX

HONORS & AWARDS

2026 **PKU International Students Academic Award**, Peking University.
Recognition of outstanding academic achievements during 2023 - 2025.

2023 - 2026 **Chinese Government Scholarship**, China Scholarship Council.
Full funding for the M.Sc. program at Peking University.

2020, 2021, 2022 **Jiangsu Provincial Outstanding Student Scholarship**.
Awarded for academic performance in three consecutive years.

2020 **Entrepreneur Scholarship**, Jiangnan University.
One-time entrepreneurship grant.

2017, 2018 **National High School Programming Contest Champion, Bangladesh**.
Two-time champion, representing Edward College.

VISION & MOTIVATION

Human intelligence has fascinated me since childhood, especially how we learn and how learning can be improved. This curiosity led me to deep learning and to questions about efficiency, supervision, and training. My recent work explores how more informative learning signals can help models make better use of their capacity. During a PhD, I hope to pursue meaningful research in deep learning or deep reinforcement learning, exploring new questions and ideas and remaining open to directions beyond my current work.

RESEARCH PROJECTS

TreemapMix: Multi-Image Augmentation with Probability and Ordinal Supervision

NeurIPS 2026 • *2026 - Present*

Conceived and led TreemapMix, extending my work on ordinal supervision from teacher-derived rankings to supervision generated directly from multi-image composition. Developed a Dirichlet-controlled K-way augmentation using area-proportional treemap layouts and cyclic views, with soft cross-entropy for probability supervision and a Plackett–Luce objective for ordinal supervision. Developed the mathematical formulation, implementation, and large-scale experimental framework.

PLD: A Choice-Theoretic List-Wise Knowledge Distillation

NeurIPS 2025 • *Sep 2024 - Sep 2025*

Asked whether knowledge distillation requires matching a teacher's exact probability distribution, and formulated class ranking as an alternative supervisory signal. Developed a **choice-theoretic Plackett–Luce list-wise objective** that integrates with CE+KL distillation pipelines and demonstrated ranking-based matching as an effective alternative to KL minimization.

Plackett–Luce Supervision for Knowledge Distillation

Peking University M.Sc. Thesis • *2024 - 2026*

Extended Plackett–Luce supervision to **multi-teacher knowledge distillation**, jointly transferring the distinct class rankings produced by multiple teachers. Investigated how ordered class preferences can serve as supervisory signals for transferring knowledge.

DecentFL: Decentralized Federated Learning

IEEE Internet of Things Journal • *2026*

Contribution: experimental implementation and benchmarking. Implemented pipelines for FedAvg, FedProx, FedDBE, and GPFL. Evaluated CNN and ResNet-18 models across 100 non-IID clients under Dirichlet and pathological label-skew settings. Conducted multi-seed experiments over 100–200 global rounds to evaluate convergence behavior under blockchain-coordinated decentralized training.

A Self-Instructing Language Model for Multi-Round Conversation

Bachelor's Thesis • *Sep 2022 - Jun 2023*

Developed a **self-instruction pipeline for multi-turn conversational data**, including data processing and model training. Fine-tuned LLaMA-1 7B for conversational instruction following and released the model and Refined ShareGPT dataset on Hugging Face.

TECHNICAL PROJECTS

Policy Optimization for Language Models

PKU Reinforcement Learning • *Mar 2024 - Jul 2024*

Studied preference learning and policy optimization for language models. Released experimental code and model checkpoints on Hugging Face: policy-optimization-lm.

YE-pop: Synthetic Multimodal Dataset

PKU Computer Vision • *Aug 2023 - Dec 2023*

Built YE-pop, a synthetic dataset of **500K images** for multimodal training. Designed rendering and annotation pipelines and released the dataset on Hugging Face, where it has been used for downstream diffusion-model training.

BEYOND RESEARCH

Yapping with LLMs | Watching movies | Swimming | Hiking & travel | Table tennis | Badminton